MMT Lemma 6 in the Jacobian-transfer screen

2026-10-13 · research/jacobian_transfer_lemma6_20261013/ · PR #1772 (aburan28/crypto)

Question

Take a binary curve $E : y^2 + xy = x^3 + ax^2 + b$ over $K = \mathbb{F}_{2^N}$ with $\mathrm{Tr}(a) = 1$ and an even relative degree $n = N/l$. What are the GHS magic number and the descent route over $k = \mathbb{F}_{2^l}$? Before this change the screen (`src/cryptanalysis/jacobian_transfer/`) did not answer that question. It priced two heuristic rows, m and $m+1$, and marked the curve `unresolved` when they disagreed.

Answer (derived; exact on each curve)

- a never changes the magic number $m = \dim \mathrm{Span}\left\{\left(1, \sigma^i\left(\sqrt{b}\right)\right)\right\}$ (Maurer–Menezes–Teske, LMS J. Comput. Math. 5 (2002), eq. (5)).
- a decides whether the descent exists. MMT Lemma 6, condition (8): n odd, or $\mathrm{Tr}(a) = 0$, or $\mathrm{Tr}_{K/k}(b) \neq 0$.
- Why: when n is even and $\mathrm{Tr}_{K/k}(b) = 0$, the sum of the n Artin–Schreier conjugates of $z^2 + z = x + a + b/x^2$ reduces to the constant $\mathrm{Tr}_{K/k}(a)$. Its $\wp$-root lies in $\mathbb{F}_{q^2}$, and lies in k only when $\mathrm{Tr}(a) = 0$. Otherwise every lift of the q-Frobenius has order $2n$, and there is no descent.
- The screen now gives one exact row per tower; a row where (8) fails is `descent_obstructed`.

The full derivation and the list of changed code are in `docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md`.

What changed

Sources:

- **Before:** `docs/curves/jacobian-transfers.json` at commit `b8ce3f697`.
- **After:** the same file at this report's commit.
- Registry SHA-256 `9e420a7f…1219e872`, `--budget 200000`.

The values are copied by `extract_rows.py`. Magic numbers, genera and condition (8) are exact invariants. The ratio $\log_2(\mathrm{rho}) - \log_2(\mathrm{route})$ is a formula under the frozen cost model, not a measurement.

curve	r, bits	GF(4) row before	GF(4) row after	verdict before → after
c2pnb176w1	160.0	$m = 8$ lower bound, heuristic, +23.64; $m = 9$ upper bound, −5.76	$m = 8$ exact, genus 128, (8) holds, +23.64	unresolved → formula_advantage
c2pnb272w1	256.0	$m = 8$ lower bound, heuristic, +72.20; $m = 9$ upper bound, +41.32	$m = 8$ exact, genus 128, (8) holds, +72.20	formula_advantage → formula_advantage
c2pnb304w1	288.0	$m = 8$ lower bound, not carried; $m = 9$ upper bound, +57.56	$m = 8$ exact, genus 128, not carried (256 < 288 bits)	unresolved → formula_marginal (GF(2^{76}), $m = 4$, genus 8, +2.00)

ICV1 slugs:

- c2pnb176w1: `icv1-f2m176-t381619313747976117155959 47-5639a6d4`
- c2pnb272w1: `icv1-f2m272-t1141827078204758669404223296143342659589 71-985955bd`
- c2pnb304w1: `icv1-f2m304-tm194905006687678203715544824401794033657667 2053-02fbe089`

Across all 2091 registry curves, these three are the only verdict or GF(4) row changes.

The isogeny-extension rows of these three curves are gone. For $\mathrm{Tr}(a) = 1$ and even n they now count only targets that satisfy (8), and none remain below the direct m. Every removed row was `not_carried` or `boundary`.

Condition (8) fails on one registry curve, `icv1-f2m18-t999-40751283` ($b \in \mathbb{F}_2$, $\mathrm{Tr}(a) = 1$), at $l = 9, 3, 1$. Its rows were already not carried, and its verdict stays `boundary`.

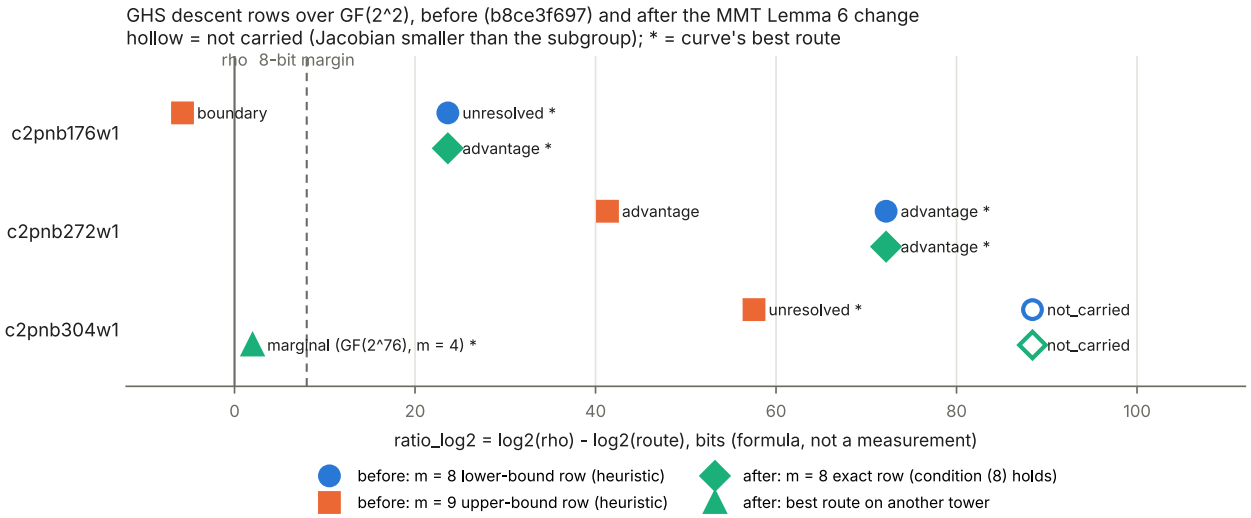


Figure 1: GHS rows over GF(2^2) before (`b8ce3f697`) and after, plus c2pnb304w1's new best route (`gf4_rows.svg` from `gf4_rows.csv`). Hollow markers are not carried; * marks each curve's best route. The dashed line is the 8-bit advantage margin.

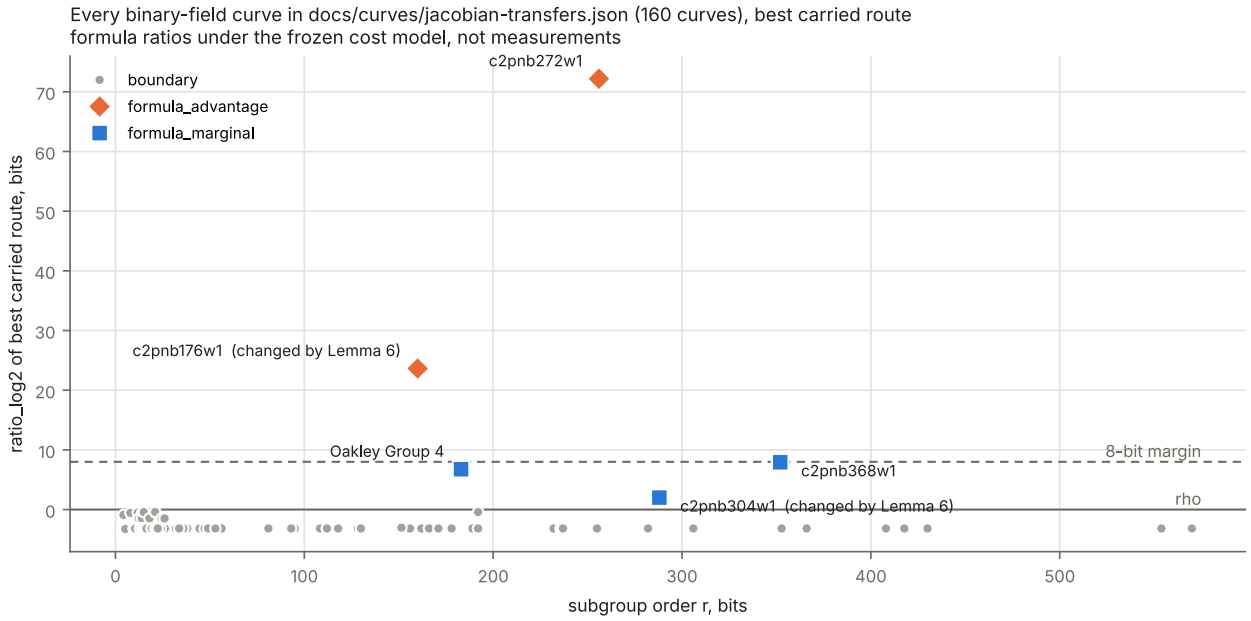


Figure 2: Best carried route of every binary-field curve in the file (160 curves): 155 `boundary`, 3 `formula_marginal`, 2 `formula_advantage` (`binary_curves.svg` from `binary_curves.csv`). The two curves whose verdict this change moved are labeled.

Checks run

- `cargo test --release --lib`: 4065 passed, 0 failed, 114 ignored.
- `mmt_condition_8_matches_the_relative_trace_of_b`: exhaustive over $\mathbb{F}_{2^{12}}$ on all six towers. It compares condition (8) with $\sum b^{q^i}$ computed directly and checks the kernel size $q^{n-1}-1$. It also checks that m and the type do not depend on a, and that `ec_trapdoor::mmt_condition_8` agrees.
- `ec_trapdoor` tests: `mmt_condition_8_counts_n12`, `audit_curve_puts_descending_towers_first`, `trapdoor_curves_satisfy_condition_8`.
- `jacobian_transfers check` and the `JACOBIAN_TRANSFERS.md` table: current.

Scope and limits

- These are formula-level prices on exact invariants. No cover is constructed, no map's action on the subgroup is verified, no relation is collected, and nothing is timed.
- No speedup, key-recovery or deployed-curve security claim is made. The `formula_advantage` rows are leads with every obligation in `docs/endomorphism-rules.md` open.
- Condition (8) is used as MMT state it. The exhaustive test checks this implementation against the definition; it does not re-prove the lemma.

Graphs checked

- The 2026-10-10 scan's `chart.svg` and `report.pdf` are a dated 321-curve snapshot. They are left unchanged, and its README has a dated correction note pointing here.
- The registry-wide `jacobian_transfers chart` labels every advantage and marginal curve (846 + 47 of them toy extension-field curves), and those labels overlap at 2091 curves. Figure 2 is drawn from the same file instead.